

```

INCLUDE Irvine32.inc
.data
myByte BYTE 20h
myWord WORD 1234h
myDword DWORD 12345678h
.code
main PROC
mov esi, offset myByte
mov eax, esi
call writehex
call crlf

mov esi, offset myWord
mov eax, esi
call writehex
call crlf

mov esi, offset myDword
mov eax, esi
call writehex
call crlf

mov ax, WORD PTR [myDword + 2]
movzx eax, ax
call writehex
call crlf

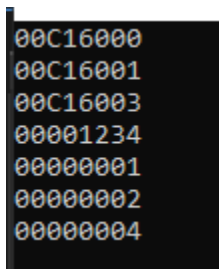
mov bx, TYPE myByte
movzx eax, bx
call writehex
call crlf

mov bx, TYPE myWord
movzx eax, bx
call writehex
call crlf

mov bx, TYPE myDword
movzx eax, bx
call writehex
call crlf

exit
main ENDP
END main

```



```

00C16000
00C16001
00C16003
00001234
00000001
00000002
00000004

```

```

INCLUDE Irvine32.inc

```

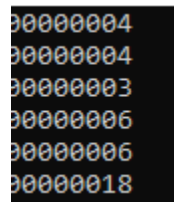
```
.data
arr1 BYTE 60,70,80,90
arr2 WORD 500h,600h,700h
arr3 DWORD 6 DUP(0)
```

```
.code
main PROC
mov ax, lengthof arr1
movzx eax, ax
call writehex
call crlf
mov ax, sizeof arr1
movzx eax, ax
call writehex
call crlf
```

```
mov bx, lengthof arr2
movzx eax, bx
call writehex
call crlf
mov bx, sizeof arr2
movzx eax, bx
call writehex
call crlf
```

```
mov cx, lengthof arr3
movzx eax, cx
call writehex
call crlf
mov cx, sizeof arr3
movzx eax, cx
call writehex
call crlf
```

```
exit
main ENDP
END main
```



A screenshot of a debugger's memory window showing a list of memory addresses and their corresponding values. The addresses are 00000004, 00000004, 00000003, 00000006, 00000006, and 00000018. The values are 00000004, 00000004, 00000003, 00000006, 00000006, and 00000018 respectively.

```
INCLUDE Irvine32.inc
.data
arrayB BYTE 15h, 25h, 35h
arrayW WORD 4444h,5555h,6666h
```

```
.code
main PROC
mov esi, OFFSET arrayB

mov al, [esi]
```

```
movzx eax, al
call writehex
call crlf

;
inc esi
mov al, [esi]
movzx eax, al
call writehex
call crlf

inc esi
mov al, [esi]
movzx eax, al
call writehex
call crlf

mov esi, OFFSET arrayW

mov ax, [esi]
add esi, 2
movzx eax, ax
call writehex
call crlf

mov ax, [esi]
add esi, 2
movzx eax, ax
call writehex
call crlf

mov ax, [esi]
movzx eax, ax
call writehex
call crlf
```

```
exit
main ENDP
```

```
00000015
00000025
00000035
00004444
00005555
00006666
```